# Favorita Grocery Sales Analysis

## Stakeholder Business Report

**Project:** Corporación Favorita Grocery Sales Analysis
**Report Type:** Business Intelligence & Exploratory Data Analysis
**Analysis Period:** January 1, 2013 – August 15, 2017
**Dataset Size:** 125.5 Million Records
**Store Network:** 54 Stores
**Product Catalog:** 4,100 Products
**Primary Tools:** Python, Polars, Power BI

---

# Executive Summary .1

## Project Overview

This report presents the findings of an exploratory business analysis of Corporación .Favorita's grocery sales data, covering the period from January 2013 through August 2017

The objective was to investigate sales performance across stores, products, geographic locations, and time periods, while examining selected external and operational factors .associated with observed sales patterns

The analysis was designed to provide a descriptive understanding of the business's sales structure, identify areas of concentration and variation, and highlight patterns that may .support further investigation and business decision-making

Sales performance is primarily measured using **units sold**, rather than monetary revenue or .profit

## Key Business Findings

**.Sales are concentrated among a relatively small group of stores .1**

The ten highest-volume stores accounted for approximately 40.17% of total recorded unit .sales, representing about 18.5% of the store network

Store 44 recorded the highest cumulative sales volume, with approximately 62.09 million .units

**.Store types exhibit substantially different sales profiles .2**

Type A recorded the highest aggregate and average unit sales per store, with approximately .353.04 million units across nine stores

Type D generated a comparable aggregate volume of approximately 351.06 million units .across 18 stores

The difference between these two types highlights the importance of considering both .network size and average store performance

**.Product sales are concentrated within a limited number of product families .3**

.Grocery I and Beverages accounted for approximately 52.20% of total recorded unit sales

Grocery I alone contributed approximately 31.99%, making it the largest product family by .unit volume

**.Sales patterns exhibit clear temporal variation .4**

December recorded the highest average normalized monthly sales index, approximately .134.97, indicating sales levels around 35% above the average month

.Sunday and Saturday also exhibited higher average daily sales than other weekdays

**.Sales growth cannot be explained by store network expansion alone .5**

A fixed-store comparison involving the same 46 stores showed an approximately 128.3% .increase in unit sales between January 2013 and July 2017

This suggests that the observed growth was not simply the result of adding more stores to .the comparison group

**.Promotional records were associated with higher average unit sales .6**

Across 2015–2017, promotional observations generally recorded higher average unit sales .than non-promotional observations

The relationship remained visible in comparisons conducted within the same store, product, .year, and weekday

However, these findings are observational and do not establish the causal impact of .promotions

# Overall Assessment

The analysis identifies a business with considerable variation in sales across stores and products, meaningful temporal patterns, and observable associations between promotional .activity and unit sales

These findings provide a descriptive foundation for understanding sales performance and identifying areas where more targeted operational and commercial analysis may be .valuable

---

# Business Context and Analytical .2 Scope

## Dataset Overview

.The analysis was conducted using the cleaned Favorita grocery sales dataset

The dataset contains approximately 125.5 million sales records covering January 1, 2013, .through August 15, 2017

The supporting business dimensions include store attributes and product metadata, allowing .sales records to be examined across multiple business dimensions

:Key dataset characteristics

- .cleaned sales records 125,497,040
- .stores 54
- .unique products 4,100
- .product families 33
- .Five store types: A, B, C, D, and E
- .An observation period spanning more than four and a half years

## Analytical Objectives

:The project was structured around six analytical areas

**Executive Overview .1**

Provide a consolidated view of business performance and highlight the most important .findings

**Time Analysis .2**

Investigate sales trends, monthly patterns, seasonality, weekday variation, and changes over .the observation period

**Store Performance .3**

Evaluate store-level sales, rankings, contribution, year-over-year growth, store-type .differences, and performance variability

**Product Performance .4**

Identify the most significant product families and products, examine sales concentration, and .investigate product-level variation

**Geographic Analysis .5**

Explore how sales are distributed across geographic dimensions and examine differences .between locations

**External Factors & Drivers .6**

Investigate the relationship between promotional activity and observed sales, alongside .temporal and operational patterns that may help explain sales variation

---

# Time Analysis .3

## Sales Trends Over Time

The analysis examined daily and monthly unit sales to understand how sales volume .evolved throughout the observation period

.Average daily sales increased substantially between the beginning and end of the dataset

:For example

- .January 2013 recorded approximately 333,142 average daily units
- .April 2017 recorded approximately 863,177
- .May 2017 recorded approximately 868,124
- .July 2017 recorded approximately 871,338

.These observations indicate a substantial increase in recorded daily sales volume

However, aggregate growth should not automatically be interpreted as an equivalent increase in underlying customer demand. Changes in store coverage, product assortment, .availability, promotions, and other operational factors may contribute to the observed trend

## Seasonal Patterns

Monthly sales were normalized against each year's average daily sales to improve .comparability between years with different overall sales levels

:December recorded the highest average normalized sales index

- .December: 134.97
- .November: 111.15
- .September: 107.40
- .October: 106.85
- .February: 83.52

.An index of 100 represents the average month within a year

December therefore recorded average daily sales approximately 35% above the annual .monthly baseline

December was also the highest month by average daily unit sales in every complete year .from 2013 through 2016

This consistency suggests a meaningful seasonal pattern that warrants consideration in .inventory planning, workforce allocation, and promotional scheduling

## Day-of-Week Patterns

Daily sales were compared across weekdays, using annual normalization to account for .differences in overall sales levels

:The results showed

- .Sunday: 130.83
- .Saturday: 121.62
- .Monday: 96.87
- .Wednesday: 92.13
- .Friday: 90.08
- .Tuesday: 88.98
- .Thursday: 79.49

Sunday and Saturday exhibited the highest average sales levels, while Thursday recorded .the lowest

.This pattern suggests that sales activity is not evenly distributed throughout the week

From an operational perspective, these differences may be useful when reviewing staffing .requirements, replenishment schedules, and inventory availability

## Fixed-Store Growth Comparison

To investigate whether the increase in total sales could be explained by store network .expansion, the analysis compared the same 46 stores at two points in time

| July 2017 | January 2013 | Metric |
|---|---|---|
| 23.568M | 10.327M | Total unit sales |
| 46 | 46 | Store count |
| 512,349 | 224,509 | Average monthly sales per store |

.Both months contained 31 calendar days

.Sales increased by approximately 128.3% between the two periods

This comparison indicates that growth was also present within a fixed group of stores, rather .than being attributable exclusively to an increase in the number of stores

Nevertheless, the comparison does not isolate the contribution of demand, product .assortment, availability, or other operational changes

## Business Implications

:The temporal patterns suggest several areas for further operational consideration

- Inventory and replenishment planning should account for elevated December .demand
- Staffing and operational capacity may benefit from consideration of weekday sales .variation
- .Historical sales patterns can support more informed seasonal planning
- Long-term sales growth should be investigated alongside changes in product .assortment and availability

---

# Store Performance .4

## Store-Level Sales Distribution

Store performance was evaluated using total recorded unit sales across the entire available .observation period

:The highest-volume stores were

| Total unit sales | Store | Rank |
|---|---|---|
| 62.09M | 44 | 1 |

| | | |
|---|---|---|
| 54.50M | 45 | 2 |
| 50.95M | 47 | 3 |
| 50.48M | 3 | 4 |
| 43.42M | 49 | 5 |

Store 44 recorded the highest cumulative sales volume, while Store 52 recorded the lowest, .at approximately 2.70 million units

The difference between the highest- and lowest-volume stores demonstrates substantial .variation across the store network

.The top ten stores accounted for approximately 40.17% of total unit sales

This concentration suggests that a relatively small subset of stores represents a substantial .share of the network's recorded sales volume

.However, total sales volume does not directly measure operational efficiency or profitability

# Performance by Store Type

.Store types were compared using aggregate sales and average sales per store

| Average Units per Store | Total Units Sold | Number of Stores | Store Type |
|---|---|---|---|
| 39.23M | 353.04M | 9 | A |
| 18.16M | 145.25M | 8 | B |
| 10.96M | 164.43M | 15 | C |
| 19.50M | 351.06M | 18 | D |
| 14.96M | 59.82M | 4 | E |

.Type A recorded the highest total sales and the highest average sales per store

.Type D generated a similar aggregate sales volume despite having twice as many stores

.Type C recorded the lowest average sales per store

These results demonstrate that aggregate sales and average store performance provide .different perspectives on the network

# Variation Within Store Types

.The analysis also examined variation between individual stores within each type

.The coefficient of variation (CV) was calculated to compare relative dispersion

| Coefficient of Variation | Store Type |
|---|---|
| 51.29% | D |
| 44.29% | A |
| 36.53% | B |
| 36.03% | C |
| 24.82% | E |

Type D exhibited the highest relative variation, indicating that its individual stores differed .more substantially in sales volume relative to the group's average

.Type E exhibited the lowest relative variation

This does not automatically indicate superior performance. It describes the consistency of .sales volume across stores within each type

# Year-over-Year Performance

Year-over-year comparisons were conducted using aligned January 1–August 15 periods to .avoid comparing partial-year data with complete years

:Between 2013 and 2014

- .Mean store growth was approximately 41.59%
- .Median growth was approximately 41.34%
- .of 47 stores with valid comparisons recorded increased sales 46

:Between 2016 and 2017

- .Mean growth was approximately 8.81%
- .Median growth was approximately 6.58%
- .Minimum growth was -4.60%
- .Maximum growth was 36.78%

The results indicate that store growth rates varied considerably across periods and individual .locations

The IQR method identified unusual growth rates in the 2013–2014 comparison, including .Stores 24, 36, and 49

.These stores represent potential investigation areas rather than confirmed data errors

## Business Implications

:The store analysis highlights opportunities to

- .Investigate operational differences between high- and low-volume stores
- .Examine the factors contributing to variation within Type D
- .Understand the characteristics of stores with unusually high growth
- .Use store-level performance benchmarks when evaluating operational changes
- .Avoid treating cumulative unit sales as a direct measure of profitability or efficiency

---

# Product Performance .5

## Product Family Contribution

Product sales were aggregated across 33 product families to identify the categories .contributing most significantly to total unit volume

:The largest product families included

| Contribution | Total Units Sold | Product Family |
|---|---|---|
| 31.99% | 343.46M | Grocery I |
| 20.21% | 216.94M | Beverages |
| 11.43% | 122.70M | Produce |
| 9.08% | 97.52M | Cleaning |
| 6.01% | 64.49M | Dairy |
| 3.92% | 42.13M | Bread/Bakery |
| 2.97% | 31.88M | Poultry |

.Grocery I and Beverages together accounted for approximately 52.20% of total unit sales

.The seven largest families represented approximately 85.61% of total recorded unit sales

This indicates that the business's sales volume is substantially concentrated within a limited .number of product families

## Product-Level Sales Distribution

.Product-level analysis examined the distribution of sales within individual families

.Mean-to-median ratios highlighted differences in the distribution of product sales

:Examples included

- .Liquor, Wine, Beer: approximately 5.70
- .Grocery II: approximately 3.48
- .Personal Care: approximately 2.22
- .Beverages: approximately 2.02

Higher ratios suggest that a limited number of products may contribute disproportionately to .family-level unit sales

.This observation was further investigated through top-product concentration analysis

# Top-Product Concentration

The top 10% of products within each family were examined to assess their contribution to .family sales

:Examples

- .Grocery II: approximately 58.01%
- .Liquor, Wine, Beer: approximately 57.39%

These results indicate that sales concentration can exist not only between product families .but also among individual products within the same family

Families containing very few products require careful interpretation because high .concentration may be a direct consequence of their limited catalog size

# Product Performance Across Stores

Product-store sales were examined using store attributes such as type, city, state, and .cluster

.The selected analysis included 528 observed product-store combinations

This provided an additional perspective on how product performance varies across store .environments

However, an absent product-store combination cannot automatically be interpreted as zero demand or a stockout because the available sales records do not provide a fully validated .product availability history

# Business Implications

:The product analysis suggests several areas for further investigation

.Prioritize inventory monitoring for high-contribution product families ●
.Examine product-level concentration within major families ●
.Investigate whether top-selling products perform consistently across store types ●
.Review low-volume products alongside their observed selling periods and availability ●
Avoid making assortment decisions based solely on cumulative sales without ●
.considering product lifecycle and availability

---

# Geographic Analysis .6

## Geographic Performance Overview

Geographic analysis introduces the location dimension into the sales performance .assessment

By connecting sales records with store metadata, performance can be examined across .geographic attributes, including city and state

The objective is to investigate whether recorded sales volume is distributed consistently across locations or whether particular geographic areas represent a larger share of the .observed business

Geographic analysis complements store-level and store-type analysis by providing a broader .view of the network

## Analytical Considerations

:Geographic comparisons should consider

.Differences in the number of stores across locations ●
.Differences in cumulative sales volume ●
.Average sales per store within each location ●
.Differences in store types and network composition ●
.Potential differences in customer demand and product assortment ●

Aggregate geographic sales should not be interpreted as evidence that a particular location has stronger customer demand without considering the number of stores and other .operational factors

## Business Relevance

Geographic comparisons can help identify locations that warrant further operational .investigation

:Potential applications include

.Comparing sales distribution across geographic areas ●
.Identifying locations with unusually high or low recorded sales ●
.Reviewing store network composition ●
Investigating whether geographic differences are associated with store-type ●
.differences
.Supporting further location-level performance assessments ●

Geographic findings should be interpreted alongside store counts and average store-level .performance to avoid misleading conclusions based solely on aggregate volume

---

# External Factors & Sales Drivers .7

## Promotional Activity

The analysis investigated the relationship between promotional records and observed unit .sales

:The `onpromotion` field contained

.(True: approximately 7.81 million records (6.22% ●
.(False: approximately 96.03 million records (76.52% ●
.(Null: approximately 21.66 million records (17.26% ●

.Promotion labels were unavailable for all 2013 records and part of 2014

Consequently, the primary promotional comparisons focused on 2015–2017, when the .information was more consistently available

## Promotional and Non-Promotional Sales

Average unit sales per observed record were compared between promotional and .non-promotional observations

| Relative Difference | Promotional Average | Non-Promotional Average | Year |
|---|---|---|---|
| 74.9%+ | 14.69 | 8.40 | 2015 |
| 60.9%+ | 12.43 | 7.72 | 2016 |
| 84.6%+ | 13.72 | 7.43 | 2017 |

.Promotional observations recorded higher average unit sales in all three years

However, this comparison does not establish promotional effectiveness because products .selected for promotions may differ systematically from other products

# Product-Level Promotional Comparisons

Product-level comparisons were conducted to investigate whether the association was .consistent across individual products

.A minimum of 30 observations under each condition was required

:Results

- .products met the observation threshold 3,610
- .products (80.3%) had higher average sales under promotion 2,900
- .products (19.7%) had lower average sales under promotion 710
- .Median relative difference: approximately +29.20%

This indicates that higher average unit sales under promotion were common across the .eligible product population

# Store–Product Comparisons

To improve comparability, promotional and non-promotional observations were also .compared within the same store–product combination and year

| Lower Average Under Promotion | Higher Average Under Promotion | Eligible Pairs | Year |
|---|---|---|---|
| 4,003 | 11,236 | 15,242 | 2015 |
| 7,856 | 40,397 | 48,260 | 2016 |
| 3,809 | 35,149 | 38,962 | 2017 |

The comparison retained the observed association within more specific business .combinations

# Store–Product–Weekday Comparisons

.A further comparison held the store, product, weekday, and year consistent

.A minimum of 10 observations under each promotional condition was required

| Median Relative Difference | Lower Average Under Promotion | Higher Average Under Promotion | Eligible Groups | Year |
|---|---|---|---|---|
| 6.35%+ | 4,152 | 5,978 | 10,161 | 2015 |
| 34.51%+ | 9,160 | 46,767 | 55,998 | 2016 |
| 69.28%+ | 2,346 | 28,190 | 30,588 | 2017 |

The results indicate that higher average unit sales under promotion remained common even .within more narrowly defined comparisons

.Nevertheless, these comparisons remain observational

They do not fully control for promotional targeting, product availability, time-specific .conditions, or other potential confounding factors

## Other Observed Sales Drivers

.The temporal analysis also identified patterns relevant to operational planning

:These included

- .Higher average sales in December
- .Higher average sales on Sundays and Saturdays
- .Differences in sales volume across store types
- .Differences in sales performance between geographic and store-level groups

These patterns represent observed associations and should not be treated as independently .established causal drivers

## Business Implications

:The promotional and temporal findings may support further investigation into

- .Promotional planning and product selection
- .Inventory preparation during periods of elevated sales
- .Differences in promotional associations between products
- .Store-level promotional performance
- .More rigorous measurement of promotional uplift

A controlled evaluation would be required to estimate the incremental sales attributable to .promotions

---

# Recommendations for Business .8 Consideration

Based on the observed findings, the following areas merit further investigation and .consideration

## Improve Inventory Planning Around Seasonal .1 Demand

.December consistently recorded elevated sales relative to the annual monthly baseline

Historical seasonal patterns can support improved inventory forecasting and replenishment .planning

## Investigate Store-Level Performance Differences .2

The concentration of sales among the top ten stores and the substantial variation within Type D suggest opportunities to investigate differences in store operations, assortment, and .customer demand

## Prioritize High-Contribution Product Families .3

.Grocery I and Beverages represent more than half of total recorded unit sales

Their importance makes them relevant candidates for closer inventory monitoring and .product-level performance evaluation

## Evaluate Promotional Performance More Rigorously .4

Promotional observations were associated with higher average unit sales across multiple .comparison levels

A future evaluation should distinguish incremental promotional impact from the effects of .product selection, timing, and availability

## Integrate Store and Product Performance .5

Combining store-level and product-level insights can provide a more detailed understanding of where sales are concentrated and how product performance differs across store .environments

## Develop More Advanced Forecasting and Evaluation .6

.The observed temporal patterns provide a foundation for future predictive analysis

:Potential next steps include

- .Time-series sales forecasting
- .Product-level demand forecasting
- .Store-level demand segmentation
- .Promotional uplift estimation
- .Inventory optimization

These initiatives would require additional validation and modeling beyond the current .descriptive analysis

---

# Data Limitations and Interpretation .9

.The findings in this report should be interpreted in light of the following limitations

**Unit Sales Versus Financial Performance .1**

The analysis measures units sold rather than revenue, gross margin, or profit. Higher unit .volume does not necessarily indicate higher financial performance

**Negative Unit Sales .2**

.The dataset contains 7,795 records with negative unit sales

A sensitivity analysis showed that excluding negative observations changed no store .rankings, while the total adjustment represented approximately 0.0134% of net unit sales

Negative observations were retained in the primary analysis because their exact business .meaning has not been confirmed

**Incomplete Final Year .3**

The dataset ends on August 15, 2017. Comparisons involving 2017 were therefore .conducted using aligned calendar periods where appropriate

**Missing Promotion Information .4**

Promotion labels were missing for approximately 17.26% of records, limiting promotional .comparisons across the entire observation period

**Observed Sales Versus Actual Demand .5**

Recorded sales may be affected by product availability, stockouts, assortment decisions, and .promotional activity

Observed sales should therefore not be treated as a complete representation of underlying .customer demand

**Missing Store–Product Combinations .6**

The absence of a sales record does not necessarily indicate zero demand or product .unavailability

**Descriptive Rather Than Causal Analysis .7**

The identified relationships describe observed patterns and associations. They do not .independently establish causation

**Store Performance Versus Efficiency .8**

Cumulative sales volume does not account for store size, operating costs, customer traffic, .or other efficiency-related measures

---

# Conclusion .10

The Favorita Grocery Sales Analysis provides a multi-dimensional view of sales .performance across stores, products, geographic dimensions, and time

:The findings reveal several important characteristics of the observed business

- .Sales volume is concentrated among a relatively small group of stores
- Store types exhibit meaningful differences in aggregate performance and within-type .variability
- .Grocery I and Beverages account for a substantial share of total unit sales
- .Sales exhibit consistent seasonal and weekday patterns
- .Growth remains substantial when comparing a fixed group of stores
- Promotional records are consistently associated with higher average unit sales .across several comparison levels

Together, these findings establish a descriptive baseline for understanding the structure and .distribution of recorded sales

The analysis also identifies areas where additional investigation may provide greater business value, particularly in inventory planning, store-level performance evaluation, .product assortment, and promotional effectiveness

**The central outcome of this project is not simply a ranking of stores or products, but a structured understanding of how sales volume is distributed, how it changes over .time, and where further business investigation may be most valuable**

---

**Report Status:** Exploratory Analysis Completed
**Dashboard:** Six-page Power BI Business Intelligence Report
**Analysis Framework:** Descriptive and Diagnostic Analytics
**Primary Measurement:** Unit Sales